



ARCHITECTURE AND FLOWS

Architecture and Flows

System boundaries, recognition, tuning, game input, and deployment.

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A camera-to-controller system for the **VirtualGlove Controller (Arduino UNO Q)** and supported RetroArch consoles.

This guide describes the current implementation reviewed on September 18, 2026. It is a map of production responsibilities, data flows, interfaces, and failure behavior-not a chronology of experiments. The decisions and discarded paths that led here are recorded in the [Engineering Journey](#). Supported replay, tracing, and measurement commands are documented separately in the [Engineering Toolkit](#); release and deployment maintenance remains in the complete Git checkout.

Read this first

VirtualGlove observes a hand on the VirtualGlove Controller, turns its measurements into controller states, and sends those states to a virtual input device on the console. The browser configures and explains that process; it is not required in the per-frame gameplay path. The VirtualGlove Controller microcontroller drives the status matrix; Linux performs hand tracking and gesture recognition.

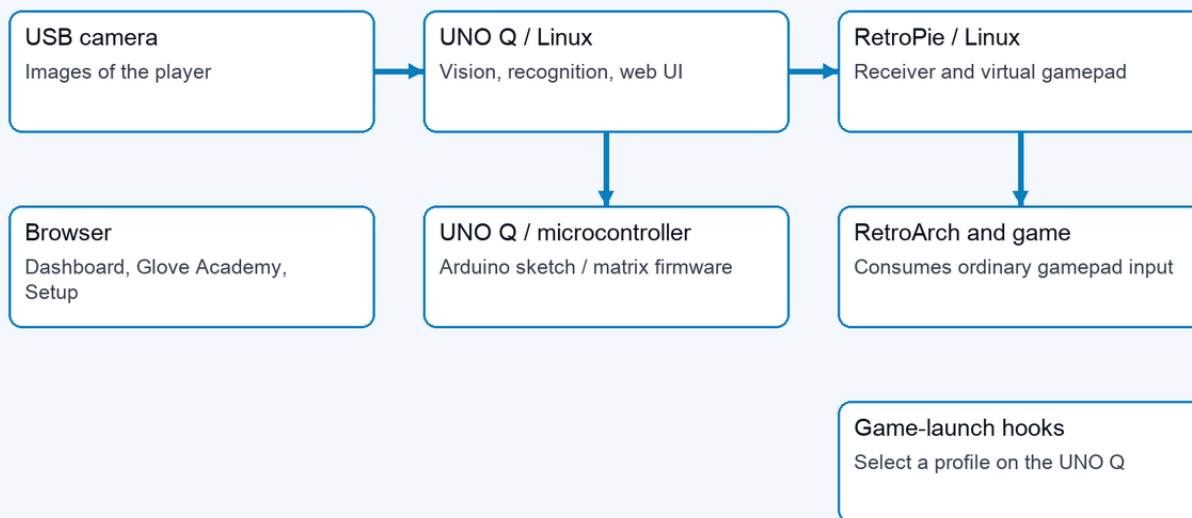
There are four independent questions: which game profile is selected, whether the camera is running, whether the player has armed controller delivery, and whether a valid registered-game or manual context currently permits packets.

Glove Academy can open the camera while the selected profile is **Gestures off**. Glove Academy and Tune both pause game input. A healthy web page does not by itself establish that the camera, receiver, or game is working.

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Firmware and application updates	Build and deployment
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System boundaries

01 / System boundaries



Browser <-> UNO web UI; game hooks -> UNO profile relay. These are separate control paths.

System boundaries: camera to UNO Linux to a supported console; separate browser, microcontroller, and game-launch paths

Boundary	Responsibilities	Does not own
Browser	Dashboard, Play, Glove Academy, Tune, Setup, Games, Help; live feedback and user commands	Authoritative per-frame recognition or gamepad output
VirtualGlove Controller Linux application	Web server, vision-worker supervision, camera tracking, calibration, thresholds, profile mapping, network sender	RetroArch button consumption
VirtualGlove Controller microcontroller	Arduino sketch, Router Bridge commands, LED matrix animations and pairing display	Camera inference or personal thresholds
RetroPie services	Receive controller packets, expose the standard VirtualGlove gamepad, signal game launches, serve paired game-registry edits	Camera processing or physical-controller remapping
Controller Router	Optionally combine EmulationStation-configured physical sources and one VirtualGlove into stable Players 1-4 for FCEUmm	Raw keyboards, mice, frontend navigation, native Super Glove Ball, or LaunchBox
Recovered RetroPie cabinet merger	Preserve the proven I-PAC/8BitDo implementation as migration reference and rollback	New generic routing after the cabinet accepts Controller Router
Recalbox integration	Run from /recalbox/share , migrate the released Player 1 record, and operate Controller Router without patching the read-only OS	Recalbox system-image files and frontend control
Batocera integration	Run as a persistent user service, consume supported game lifecycle events, and operate Controller Router	Batocera system-image files and frontend control

Boundary	Responsibilities	Does not own
LaunchBox integration	Wrap 64-bit RetroArch launches, exact-match the game registry, and publish Player 1 through a LAN-isolated loopback RetroPad beside physical XInput and real-keyboard fallback controls	LaunchBox database files and global joypad configuration
RetroArch and game	Consume virtual-gamepad input using emulator and game mappings	Glove Academy/Tune feedback

App Lab starts [python/main.py](#) in the main application container. This supervisor runs the website and starts an isolated Python 3.12 vision worker with the sole packaged MediaPipe 0.10.35 ARM64 wheel. There is no installed 0.10.18 fallback or runtime selector.

The supervisor polls worker status, updates the matrix, and retries a worker that stops. The worker's internal HTTP interface is on loopback port 8089. The public website is on 8088, with secure Setup on 8443.

On Recalbox and Batocera, Controller Router creates each enabled uinput gamepad before RetroArch starts. RetroPie installs the same subsystem disabled and keeps its ordinary separate [VirtualGlove](#) gamepad until the user explicitly saves and applies a Router configuration.

Router translates authoritative EmulationStation mappings into canonical RetroPads. A versioned document assigns several physical sources to a player, prohibits one physical source from appearing in two players, and assigns the single paired VirtualGlove to zero or one player. It resolves current event, joystick, and RetroArch indexes at boot and before FCEUmm launch.

Buttons use source-aware hold sets. The most recently activated physical source owns an axis until neutral, then another still-held physical source resumes. Physical axes outrank VirtualGlove. Only Player 1 carries the physical hotkey; VirtualGlove Select and Players 2-4 cannot assert the hotkey-enabler.

The merged gamepads remain neutral outside FCEUmm, so EmulationStation continues to use only the original controller. Disconnect releases only physical state; the saved stable identity reconnects without relying on an event number.

On LaunchBox, the authenticated receiver translates ordinary profiles into RetroArch Network RetroPad button transitions sent only to [127.0.0.1](#). RetroArch combines that Player 1 state with its existing XInput and real-keyboard bindings. The installer chooses a random dynamic port, enables it only in the managed FCEUmm append configuration, and blocks LocalSubnet access with a program- and port-scoped Windows Firewall rule. Super Glove Ball uses a separate append configuration with Network RetroPad disabled and consumes only the guarded native-state record.

The supervisor passes the private [data/device.json](#) path to the worker using `--device-config`; the token itself is absent from process arguments. Device configuration mutations are serialized and atomically replace private files. Setup content and browser actions live in [setup_web.py](#), separate from HTTP routes.

Start/Stop intent has a single pending slot and serialized delivery attempts; the supervisor retries transient failures until the worker handler acknowledges acceptance. A newer explicit request supersedes pending intent. This mechanism does not buffer camera frames or controller state, and acknowledgement does not prove emulator consumption.

Receiver socket timeout and last-valid-packet expiry both publish neutral native state and release the virtual gamepad.

RetroPie input topologies

The standard RetroPie topology deliberately ends at a separate Linux gamepad named [VirtualGlove](#). RetroArch can accept that device beside an already configured physical joypad, which is sufficient for an ordinary new install and does not rewrite the console's existing controller layout.

The VirtualGlove development cabinet currently retains the optional [retropie/arcade-cabinet-merger](#) reference and rollback implementation. It consumes the standard [VirtualGlove](#) device alongside two I-PAC gamepad interfaces and supported 8BitDo controllers, publishing [Arcade Merged Player 1](#) and [Arcade Merged Player 2](#). The proposed migration imports those known assignments into Controller Router and changes nothing until they pass live validation.

This component was originally installed directly at `/usr/local/sbin/arcade-gamepad-merger`. The source is project-authored rather than third-party, but it was absent from Git until the proven cabinet version was recovered and checked into the optional integration directory. It remains separate from the default installer because its exact device names, I-PAC raw codes, two-player assumptions, and cabinet hotkeys are installation-specific.

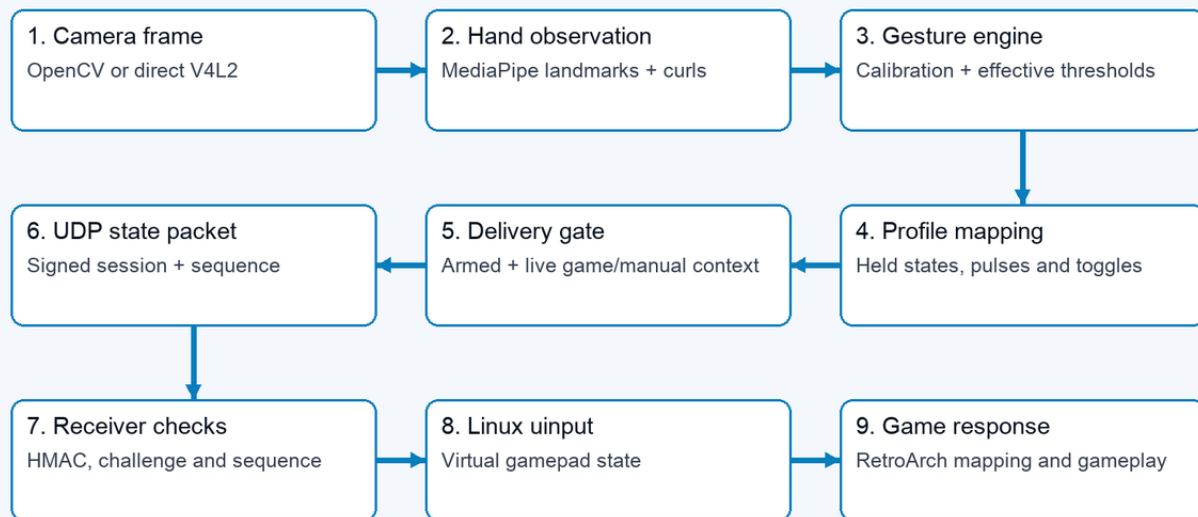
Controller Router is the shared successor for RetroPie, Recalbox, and Batocera. It generalizes the cabinet's multi-source idea to four independently enabled players, stable identities, EmulationStation mapping translation, launch-time index synchronization, exclusive FCEUmm ownership, and physical-axis priority. The older cabinet program remains available only as a tested reference and one-command rollback until the cabinet migration is accepted.

Router writes enabled Player indexes and canonical controls to FCEUmm's core-specific RetroArch override rather than the console-wide NES file. That scope is part of the native-input boundary: Nestopia (VirtualGlove) never loads the Router assignment, while the ordinary physical-controller configuration remains available to the native core.

Two app-owned support containers provide the profile-control UDP relay and local-hostname resolution. The profile relay publishes port 55356 and forwards packets to the main service without interpreting or authenticating them. The resolver connects application requests to host Avahi through private Unix sockets. These functions are kept separate from camera inference.

Camera-to-controller flow

02 / One hand movement becomes game input



Status and preview branch from the worker. Browser video is not in the controller delivery path.

Nine-stage flow from a camera frame to the game response

1. The camera layer opens a UVC capture source. The compatible path uses OpenCV. An opt-in direct V4L2 reader can instead dequeue the newest 640×480 MJPEG driver buffer and its timestamp, with automatic OpenCV fallback. A dedicated capture thread publishes only the newest frame; older unprocessed frames are superseded rather than queued.
2. MediaPipe Hands 0.10.35 runs its complete graph with four XNNPACK CPU threads. The graph first uses its palm detector, then the landmark model produces 21 hand points and normally continues from the prior hand region. The production landmark-tracking gate is 0.35 and the palm-detection gate is 0.45. Palm detection runs again when landmark tracking cannot continue. The tracker validates finite, non-collapsed palm geometry and produces a [HandObservation](#) using the selected frame's capture timestamp. MediaPipe's score is labelled as handedness certainty rather than position confidence.
3. The gesture engine compares that observation with the saved neutral calibration and effective thresholds. Directions are relative to the calibrated palm; apparent hand-size change supplies forward/backward movement.
4. Shared activation/release states and held menu poses feed the selected Programs 1-14, Programs A-I, or game-specific profile mapping. Numeric mappings can replace positional movement with depth, wrist, finger-tread, compound, or intentionally neutral output. Per-game rapid A/B exceptions arrive with the authenticated game lease. The result is a [ControllerState](#), including buttons, D-pad, axes, finger values, events, sequence, and tracking/calibration metadata.
5. The worker sends the state only if controller delivery is armed, a live registered-game lease or intentional manual Dashboard context exists, and neither practice nor tuning is active.
6. The sender establishes a receiver-issued challenge, then sends bounded HMAC-SHA256 controller datagrams over UDP 55355. Packets contain session and sequence identifiers, never the shared token.
7. The receiver checks the message HMAC, live challenge, peer, and increasing sequence. For Super Glove Ball it publishes native state before updating the unrelated virtual gamepad; other profiles preserve virtual-gamepad behavior. It creates the real virtual controller when the first accepted packet arrives.
8. Linux [uinput](#) exposes the virtual gamepad. Standard RetroPie consumes it as a separate controller; the optional RetroPie arcade-cabinet merger can first combine it with I-PAC and 8BitDo sources. Recalbox and Batocera publish their separately implemented merged Player 1. RetroArch then applies its configured input mapping before the game consumes the state.

Program 2 also derives transient centering feedback, Program 13 leaves the camera D-pad neutral for the merged physical controller, and Program 14 closes vision and neutralizes all VirtualGlove output. These are mapping and lifecycle decisions; none rewrites player calibration or recognition tuning.

The [Gameplay Guide](#) documents every numeric Program's gesture priority, compound-action release rule, and indexed game assignment.

One-second maintenance challenges provide receiver liveness. After three seconds without an authenticated reply-or immediately when the saved hostname has no usable address-the sender broadcasts only a signed hello to fresh physical-link broadcast addresses supplied by the host sampler. A valid challenge identifies the paired receiver's current IP; controller state remains unicast and newest-sample-only.

Connection cadence, safety, and load

The recurring network work is intentionally split between the time-critical controller path and slower recovery or status paths. None of the background checks waits inside MediaPipe inference. An established sender publishes the newest controller state before it performs handshake maintenance.

Activity	Normal cadence	Failure or recovery boundary	Gameplay effect and representative load
Controller state, Controller to console UDP 55355	Every fresh inference result, normally 15-30 Hz	The console neutralizes its VirtualGlove source and native state after 250 ms without a valid packet; a Recalbox/Batocera merger separately retains current physical-controller state	Time-critical path. A representative signed state is about 610 bytes, or about 18 KiB/s at 30 Hz. The timeout adds no normal-play delay.
Signed controller handshake, UDP 55355	Every 250 ms until challenged; every 1 second after establishment	Three seconds without an authenticated reply starts paired-console discovery; discovery repeats every 2 seconds until a valid challenge arrives	Runs after the current state when established. A representative signed hello is about 212 bytes; no controller state is broadcast or replayed.
Registered-game profile renewal, console to Controller UDP 55356	Every 2 seconds while RetroArch and the platform's session ownership remain active	Each renewal carries a 6-second lease. A request may try up to three 0.4-second acknowledgement waits, outside game launch. If the saved destination fails, a signed discovery request finds and briefly caches the paired Controller's current address.	Keeps the correct ROM/core mapping active and lets the Controller recover after a restart or DHCP change. A representative renewal is about 365 bytes every 2 seconds.
Setup console check, Controller to console TCP 55358	Visible Setup polls every 5 seconds; the Controller starts at most one real probe every 10 seconds	A result becomes stale after 30 seconds; destination or key changes invalidate it immediately	Status only. It does not confirm that an emulator consumed input and does not run in the inference path.
Physical-link sampler, Controller host	Every 5 seconds	Its small record expires after 15 seconds	Supplies Wi-Fi/Ethernet status and safe directed-broadcast addresses. It does not scan networks, send controller data, or affect inference.

These defaults make normal traffic small: controller states account for almost all of it, at roughly 18 KiB/s at 30 Hz, while handshake and profile maintenance add well under 1 KiB/s in steady play. A state packet is never held for a heartbeat. The 250 ms receiver timeout is a stale-input safety limit, the one-second handshake is receiver-restart detection, and the two-second profile renewal is game-context ownership; they solve different problems and should not be added together as movement latency.

With an unchanged address, a restarted receiver normally re-establishes its challenge on the next one-second maintenance exchange. With a DHCP address change or failed `.local` resolution, authenticated discovery begins after about three seconds; a lost discovery exchange can add the two-second retry interval. The next registered-game renewal can restore game context within two seconds, followed by the existing one-second initialization guard. These are recovery windows after a failure, not steady-state input delay.

Native Super Glove Ball performs MediaPipe landmark recognition synchronously. The Dashboard retains the normal hand skeleton and landmark annotation. Each fresh, geometry-valid palm observation is clamped to player reach and passed directly through **Latest coordinate** during continuous tracking. The live path does not queue, predict, extrapolate, or filter inside the emulator core. The former optical-flow experiment is retained only in Git history and the former bounded speed curve remains in historical test tooling; neither is routed by the supervisor nor exposed as a live configuration.

Native coordinates use each player's calibrated center and optional asymmetric comfortable-reach spans. A missed observation shorter than `native_xy_loss_hold_ms` (180 ms by default) holds only the last X/Y position; buttons, fingers, depth, roll, and digital directions release on the first missed native observation without inheriting the X/Y hold.

After that brief gap, Latest immediately accepts a strongly aligned forward measurement. One contradictory or unusually distant non-forward result instead holds the last reliable point until the next fresh measurement confirms the location. The guard never invents a forward coordinate or smooths normal motion. Longer tracking loss or stale input neutralizes the native sample and clears the coordinate history.

Final sustained attribution measured ordinary landmark continuation near 37 ms median and palm detection/reacquisition near 105 ms median. Two-buffer direct V4L2 diagnostics sustained approximately 30 camera dequeues per second, with MJPEG decode near 5.4 ms median and post-graph work near 2.9 ms median. Inference run-queue wait was negligible.

These figures establish the palm detector as the remaining CPU latency tail; they do not justify another smoothing layer, frame queue, or speculative parallel tracker. OpenCV with thread isolation remains the live-tested production reader. Its saved one-buffer request remains independent of the direct-reader diagnostic.

Digital FCEUmm directions instead classify every fresh hand position in a 3×3 grid anchored to the saved calibrated neutral palm position. The center square releases all positional directions, its four side regions produce cardinals, and its four corner regions produce diagonals.

Setup's **Joystick dead zone** saves the square's chosen full-frame width and height per player. Its effective size is at least 1.5 times the saved calibrated palm size. The full square translates inward at camera edges rather than clipping; live hand size and neutral jitter do not change it.

The dead zone does not alter native reach, finger gestures, or game mappings. Re-centering clears saved reach spans because they belong to the old center.

The worker sends authenticated controller state immediately after recognition. Its tuning configuration and pause gates are captured before inference, so no tuning or Dashboard lock sits between completed inference and UDP transmission. For established sessions, the gameplay state precedes periodic handshake maintenance; reply decoding is bounded per frame. UI-visible control transitions are then submitted immediately to a separate latest-only publisher.

When the browser's **Show statistics** preference is enabled, routine derived gesture and controller detail refreshes at about 10 Hz and rolling percentile summaries at 2 Hz; hidden statistics do not incur that work. Optional Dashboard statistics are off by default and browser-local. When disabled, the page does not read, render, retain, or request detailed controller and event fields.

Browser video is submitted at most five times per second and only while a stream consumer is connected. MediaPipe always uses the same fused frame preparation whether the preview is open or closed. A separate single-slot worker mirrors the display copy, draws normalized landmarks, downsizes the gameplay preview to 320×240, performs JPEG encoding, and discards a superseded preview instead of delaying gameplay.

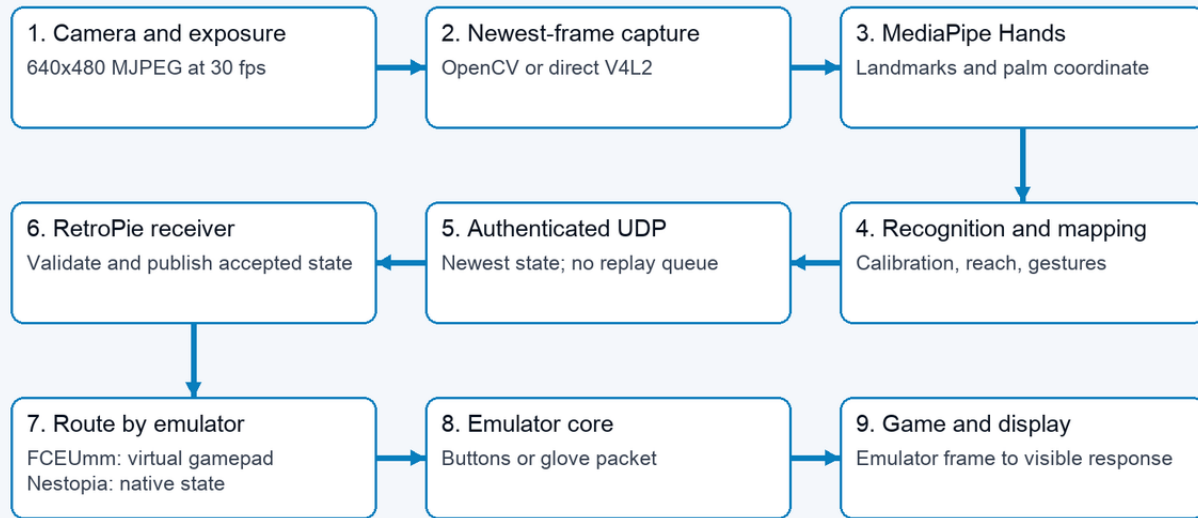
Detailed joint and landmark diagnostics follow that preview cadence; finger geometry itself is calculated once for recognition. Controller sending occurs before optional preview work, so the browser refresh rate is not the controller state update rate. Capture age, inference cadence, skipped frames, preview cost, and send time expose the local stages; none alone is an end-to-end camera-to-game latency measurement.

The shipped graph runs on the UNO Q CPU. OpenGL/OpenCL, MediaPipe Tasks GPU, MNN/Vulkan, and ncnn comparisons did not beat this complete CPU graph while preserving recognition behavior, so they remain engineering research rather than selectable gameplay runtimes. The model graph is the connected palm- detection and landmark-computation pipeline; it is broader than either neural network alone. Research details and promotion gates are consolidated in [Engineering Journey](#).

Optional manual exposure and gain are applied through the Direct V4L2 stream's existing file descriptor after the camera advertises compatible controls. This avoids a second camera opener racing the capture worker. Requested and read-back values, supported ranges, fallback, and error state are published separately. Closing the stream restores automatic exposure; the saved preference remains available for the next vision session.

Shared front end and emulator paths

08 / Camera to game: one front end, two delivery paths



The browser preview and statistics observe the worker; neither belongs in the controller delivery path.

End-to-end flow from camera and MediaPipe through authenticated delivery to the FCEUmm and Nestopia game paths

The camera, newest-frame capture, MediaPipe model, calibration, and gesture recognition are shared. The split occurs on the paired console after an authenticated packet is accepted. FCEUmm receives ordinary directions and buttons through a platform-appropriate RetroPad path. Super Glove Ball's custom Nestopia core instead reads the newest 64-byte native state and assembles the ten-byte Power Glove packet used by the ROM. Both finish in the emulator, game logic, video path, and physical display. Dashboard preview and optional statistics branch from the Controller worker and never sit between recognition and delivery.

Joystick-mode ownership by platform

RetroPie exposes VirtualGlove as its own Linux gamepad. LaunchBox keeps the physical XInput controller unchanged and adds VirtualGlove through a loopback Network RetroPad. Recalbox and Batocera need a different boundary because their launchers generate RetroArch Player 1 from the controller selected in EmulationStation.

On Recalbox and Batocera, one persistent Controller Router service creates the enabled **VirtualGlove Merged Player 1-4** outputs before gameplay. The initial installation preserves the selected Player 1 arrangement; Setup can later assign several configured physical sources to a player and the one VirtualGlove to exactly one slot. The record keeps authoritative EmulationStation mappings, including SDL indices used only to translate them into Linux event controls; those namespaces are not interchangeable on controllers whose Start, Select, or Home buttons use keyboard-class event codes.

The service never treats `/dev/input/eventN`, `/dev/input/jsN`, or a RetroArch joypad index as identity. It resolves the current event device from the saved hardware identity and resolves the merged RetroArch index from the current udev joystick order. Batocera repeats the index synchronization in its `gameStart` hook; the persistent service also detects later enumeration changes.

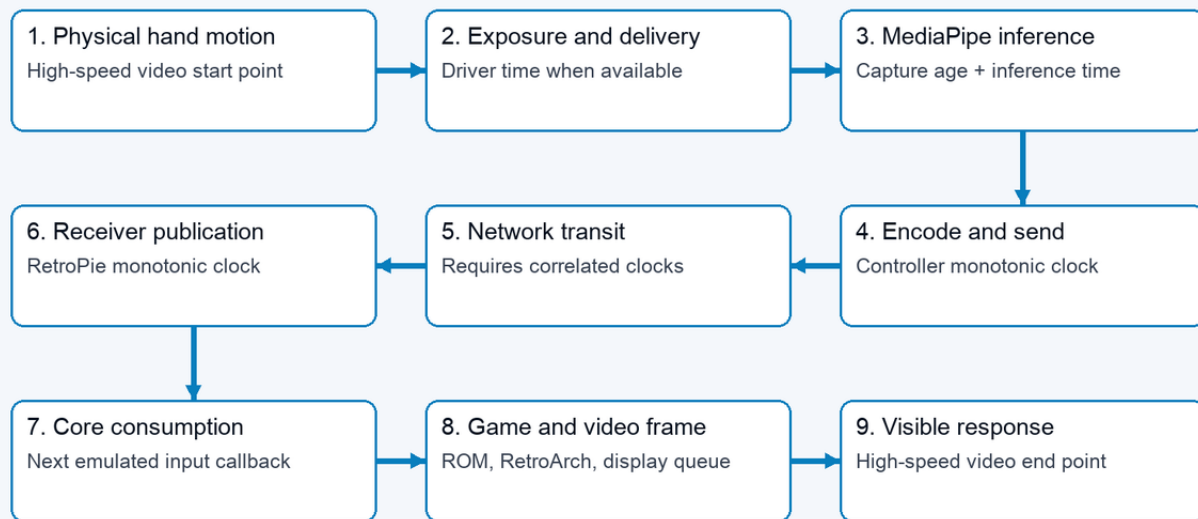
The original physical controller owns EmulationStation. When RetroArch starts, the merger uses `EVI0CGRAB` to take exclusive ownership of that controller's event stream, clears any retained physical state, and publishes only the canonical merged device. This prevents the same Select or hotkey press from reaching RetroArch, the platform hotkey service, and

the merger independently. At game exit, it publishes neutral state before releasing the grab, allowing the original controller to resume frontend navigation.

Physical directions and axes take priority while active; ordinary physical and VirtualGlove buttons combine. The physical hotkey is carried on a dedicated merged button. VirtualGlove Select can emit only the canonical Select button and cannot enable RetroArch hotkeys. A disconnect releases only physical state, so VirtualGlove can remain active until the saved controller returns.

Measurement boundaries

09 / Where end-to-end time is measured



One hand-and-screen recording measures the whole path. Software traces explain only their own clock domains.

Nine measurement boundaries from physical hand movement through camera, MediaPipe, network, emulator, and visible response

The camera experiments measure only the beginning of this chain. A valid V4L2 driver timestamp can describe driver-to-userspace dequeue time, while an OpenCV read-completion timestamp begins later and cannot reveal exposure or upstream camera buffering. Capture age and inference time describe the Controller; receiver publication and core consumption use the RetroPie's clock. Only one high-frame-rate recording containing both the real hand and the game display measures the complete visible response without synchronizing those computers.

The tests deliberately retain these boundaries. A faster camera dequeue does not prove faster recognition, a successful UDP send does not prove receiver or game consumption, and a headless ROM response excludes RetroArch presentation and display delay. The [Engineering Journey](#) records which part of this diagram each experiment actually covered.

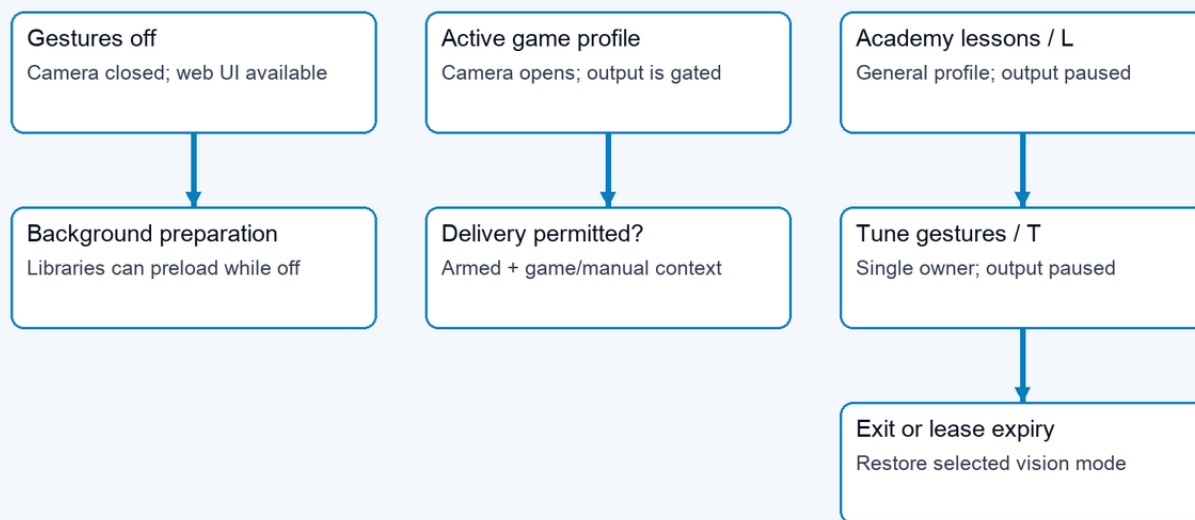
The read-only [scripts/measure-vision-status.py](#) collector deduplicates observed inference timestamps and capture sequences. Public status is cached by the supervisor, so even frequent polling observes only a subset of results. The collector separates changing profiles, preview state, and delivery conditions; it does not average overlapping rolling percentiles. Camera exposure, network reception, receiver processing, native core pickup, and physical display delay require separate evidence. Repeatable diagnostic workflows are retained in the [Engineering Toolkit](#).

The standard transport is joystick-mode gamepad emulation. Bad Street Brawler maps Glove Zap to a 180 ms simultaneous Left + Right pulse on each push activation; its FCEUmm game-specific options allow that combination. The

receiver already transports both directions. This action does not require native glove packets. Preserved finger and analogue values do not establish native original-Power-Glove support in the emulator. That remains a separate native-mode integration concern. The player-facing distinction is explained in [VirtualGlove Input Modes](#).

Runtime modes

03 / Camera activity and controller delivery are separate



Start/Stop is sticky. Game leases expire safely; Academy and Tune always pause output.

Runtime modes distinguish camera activity from the controller delivery gate

The worker keeps a lightweight control loop alive while vision is idle. Libraries can preload in the background without opening the camera. Selecting an active profile or opening Glove Academy requests camera/tracker initialization. Slow opens, reads, and cleanup run asynchronously so control requests remain responsive. The website reports startup and recovery rather than treating an empty first frame as completed initialization.

Mode	Vision profile and camera	Controller delivery	Matrix
Gestures off	Camera closed; selected profile off	No gameplay states	VirtualGlove attract animation
Program 14 - physical controller only	Camera closed; selected profile remains Program 14	No VirtualGlove gameplay states; merged physical Player 1 joypad remains available	Steady 14 profile display
Active profile	Selected game profile; camera requested	Only when armed and a game/manual context is active	Ready/tracking status and profile display
Glove Academy learning	General practice profile; camera requested	Paused	Scanning L
Tune gestures	Practice with selected tuning scope and preview	Paused, including after a game-launch request	Scanning T

Glove Academy preserves the selected game profile while using a mapping-independent practice profile for its sixteen lessons, including **Glove Zap**, **Pull Back**, both wrist rolls, close hand, and menu guard. Progress is saved for each player; completing all sixteen lessons earns the Glove Master award.

Browser leases support multiple Glove Academy tabs; the last lease ending restores the selected vision mode. Leases expire after six seconds without refresh. Dashboard also clears abandoned practice sessions.

Glove Academy learning restores its prior controller intent. Tune requires an explicit start from Dashboard when finished.

The browser downloads a backup for only the selected player, usually to its Downloads folder. Restore reads that chosen file and updates the selected player after review. The Controller keeps all players in one [data/gesture-tuning.json](#) store; portable downloads are separate copies and exclude lesson progress. See [backup locations](#).

Tune has a single owning session. Exiting or losing that session discards its recordings and preview, but saved values remain. A game launch can change the selected profile during Tune without allowing game input to escape the delivery gate. Profile/mode transitions release old controls and refresh sender sessions where required so old state does not carry into a new mapping.

The L and T render through the same sketch function: a dim letter, bright scan line, and trailing glow. Eight frames advance every 160 milliseconds. The sketch also handles other status, profile, and pairing indications; matrix activity is feedback about state, not evidence of successful game delivery.

Recognition and tuning

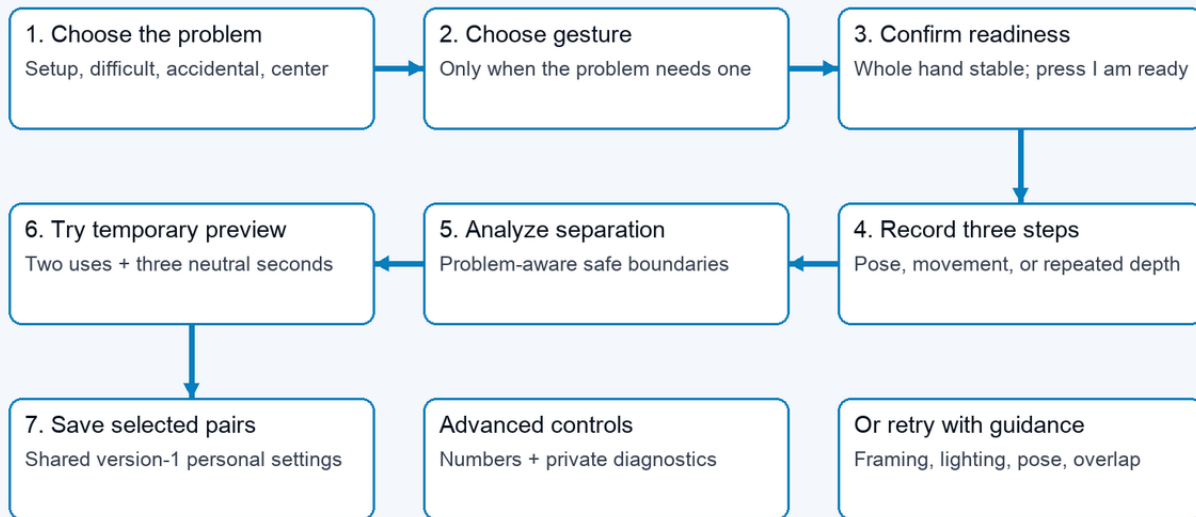
Finger curls use the strongest measured joint bend. The four fingers include the base knuckle; the thumb uses its two outer joints. The tracker prefers MediaPipe world landmarks, with an image-coordinate fallback. Recognition is threshold-based; personal setup does not retrain the MediaPipe model.

A held action has two cutoffs. Crossing **Activation** starts it; returning below the lower **Release** value stops it. This avoids flicker near one cutoff. Finger controls, direction/roll states, Glove Zap, Pull Back, and movement-based mappings use these states. Profile-specific pulses and toggles still determine what the game receives. An indicator remaining active is therefore not a promise of a continuously held game button.

V-sign and thumbs-up also require the correct extended/curled fingers and the deliberate debounce (0.50 seconds for Start and 0.15 seconds for Select), then issue a short menu pulse. Live pose feedback uses the same finger checks. Personal pairs supply the closed-finger activation and extended-finger release boundaries; untouched fingers use the existing menu defaults. A confirmed lesson can remain complete after its brief controller pulse has ended.

Directions, curls, rolls, and Closed Hand are evaluated from every fresh inference result without an additional confirmation timer. Depth actions are motion-confirmed: Glove Zap and Pull Back need two consecutive beyond-threshold observations and at least 0.10 normalized palm-scale movement in the correct direction within 250 ms. Reversal, calibration, profile transition, or tracking loss discards an unfinished candidate. Confirmed actions retain their existing hysteresis and profile-specific output semantics.

04 / Pixel Pal guides recognition personalization



Normal personalization retains no images. Controller delivery stays paused throughout the wizard.

Personalization flow: choose a problem, record guided phases, pass a preview test, and save

Tuning scope	First recording	Middle recording	Final recording
Set up a new hand	Comfortable open hand	Gentle fist, thumb curled outside fingers	Comfortable open hand
Finger/menu pose	Comfortable open hand	Selected gesture held steadily	Comfortable open hand
Glove Zap	Open hand at starting distance	Three pushes and returns	Return to starting distance
Pull Back	Open hand at starting distance	Three pull-backs and returns	Return to starting distance
Direction or wrist roll	Starting position and wrist angle	Selected movement held steadily	Return to starting position and angle

Pose, direction, and roll recordings last two seconds; the repeated depth-motion step lasts six. Recording is enabled after a calibrated, geometry-valid hand has remained completely inside the image for one second. A user-controlled two-second countdown precedes every sample.

Each step needs at least twelve accepted samples. The manager accepts calibrated, geometry-valid detected hands, rejects repeated frames and non-finite measurements, and caps samples per recording. Tracking gaps contribute no samples; too few samples require a retry. Neutral calibration changes invalidate recordings and previews.

For each adjusted component, analysis compares the 95th percentile of both open/rest phases with the performed phase. It requires a gap of at least 0.08. Standard setup places activation/release at 65%/30% of the gap; difficult and accidental paths use 55%/30% and 75%/40%. Repeated depth motion uses its upper quartile so returns to neutral are not misread as failed movement.

Hand setup observes both states for all five fingers. Individual gesture tuning can run without it. Fingers extended throughout retain their existing settings; extended-only samples cannot establish a curled boundary. Automatic suggestions now check all required fingers against the candidate configuration using the same pose checks as recognition.

At least 90% of accepted samples must match the complete pose simultaneously. The opening and release phases must also show all selected fingers extended in at least 90% of samples. A failure names the finger and phase; no suggestion is retained. Strong curls cannot compensate for fingers that should be extended.

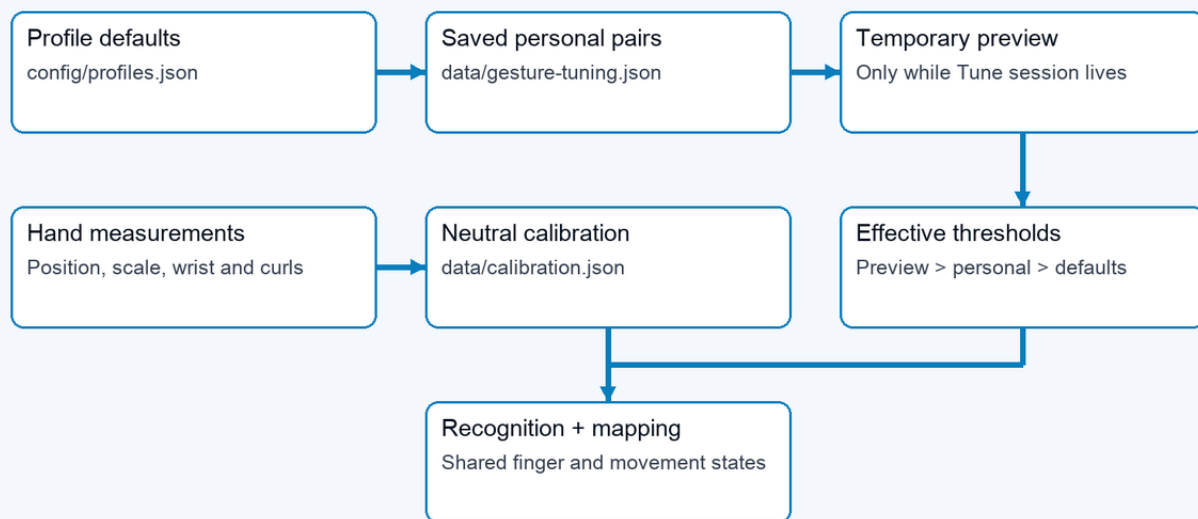
Thumbs-up checks a straight thumb and four curled fingers; it does not impose an upward screen direction. Manual threshold edits validate range and scope, not recorded pose quality. Live testing is still needed.

The candidate is temporary until the same recognition path observes two complete activation/release cycles and three neutral seconds. Only then can the wizard atomically merge selected pairs into the active player's version-6 record. Positional movement is not a gesture-tuning channel: one per-player center-box scalar drives a stateless 3×3 classification, and calibration jitter may enlarge its effective size.

Raw gesture controls remain inside Advanced. Normal personalization retains no camera recording. The separate diagnostic path deletes its temporary AVI after producing an aggregate-only report.

Profile and configuration flows

05 / Effective thresholds and calibration



Top arrows show override priority, not file writes. Neutral calibration is a separate reference.

Threshold precedence and the separate neutral-calibration reference

Effective settings are resolved component by component: shipped shared recognition defaults, then the active player's saved overrides, then temporary Tune preview. The gesture engine receives the resulting configuration during frame processing, so saved values also apply when controlling a game. Adjusting a finger changes other gestures that use that finger; it does not change the button assignments in a game profile.

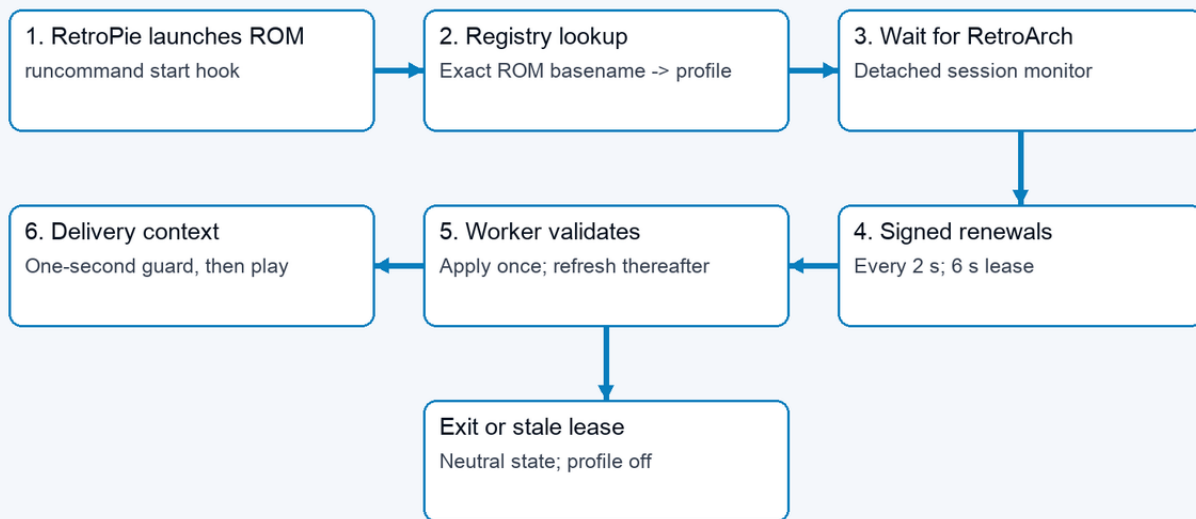
Data	Owner and lifetime	Purpose
config/profiles.js on	Shipped project source	One shared set of recognition parameters; profiles remain output mappings

Data	Owner and lifetime	Purpose
data/gesture-tuning.json	VirtualGlove Controller, persistent	Version-4 player presets, per-player calibration, sensitivity, Academy progress, and required-center flag; versions 1-3 migrate with a backup
data/calibration.json	VirtualGlove Controller, private persistent	Neutral palm position, apparent scale, wrist angle, and positional jitter for the installed camera and player
data/device.json	VirtualGlove Controller, private persistent settings	Destination, selected settings, pairing-related configuration
Tuning samples, preview, leases	Worker memory only	Temporary measurement and ownership state
config/games.json	Shipped default registry	Exact ROM-name mappings copied to the selected console installation
Console registry and launcher settings	Console, persistent	Active game-to-profile mappings and Controller destination
data/models/hand_landmarker.task	VirtualGlove Controller, verified cache	Reusable pretrained hand-landmark model

Neutral calibration is distinct from hand setup. It accepts 24 detected hand observations with finite, non-collapsed landmark geometry, centers position, depth, and roll, records 95th-percentile X/Y jitter, and lets movement thresholds rise only when needed to remain safely above that noise; hand setup establishes finger thresholds.

The app reuses valid neutral calibration across Glove Academy, profile changes, camera reconnects, and worker restarts. Recalibrate after moving the camera or changing playing position. Ordinary updates preserve [data/](#) rather than replacing it with example configuration. The portable release baseline is [config/profiles.json](#); raw neutral coordinates are deliberately machine- and player-local.

06 / A game launch selects a gesture profile



Start/Stop is stored separately. Armed output resumes only with a live game or manual context.

Profile-selection flow from the console game session through the Controller relay and worker

At game launch, the platform integration looks up the exact ROM basename. On RetroPie this uses runcommand hooks and a user-owned session marker; Recalbox uses a bounded RetroArch monitor, Batocera uses game lifecycle events, and LaunchBox uses its RetroArch wrapper. For a registered game, the integration sends a signed profile renewal every two seconds to Controller UDP 55356 while the platform-specific game lease remains valid. It derives the active core from the platform's authoritative launch or process context and includes it in each renewal.

Each renewal carries a bounded six-second lease. The app-owned relay forwards the bytes to the worker; the worker authenticates them and treats repeated renewals as lease refreshes rather than profile transitions. The acknowledgement travels back through the relay. The relay has no shared token and cannot declare a profile applied.

If the console's saved Controller address is stale or cannot be resolved, the profile sender broadcasts a signed, ROM-free discovery request to the physical IPv4 LANs. The Controller returns a signed, request-matched discovery acknowledgement. The console then sends the actual profile renewal by unicast and caches that authenticated address for 30 seconds. The cache is bounded and process-local; it neither changes `launcher.json` nor replaces pairing. A wrong key, malformed reply, or unrelated Controller cannot claim the session. This is the reverse-direction counterpart to the Controller's authenticated discovery of its paired console on UDP 55355.

The first live renewal changes profile once and starts a one-second initialization guard. A VirtualGlove Controller application restart can therefore rediscover an already-running registered game from the next renewal without exposing the runcommand menu to hand input.

Game-end hooks, RetroArch termination, marker replacement, unknown games, and lease expiry request or produce neutral/off state. The player's armed/stopped choice is stored separately: Stop remains sticky, while armed alone never authorizes output.

Manual Dashboard profile selection provides an explicit testing context without pretending that a registered game is running. Unsupported or unregistered games do not gain a mapping merely because their filenames resemble a registered title.

The worker selects native input only for the exact combination of the [super_glove_ball](#) profile and the reported [lr-nestopia-powerglove](#) core. The same ROM in FCEUmm, another core, or an unknown core uses joystick output. A reported core change is a profile transition even when the ROM profile is unchanged, so all retained recognition and controller state is cleared before the new output path becomes active. Worker status reports both [emulator](#) and the derived [input_mode](#).

Setup's Games editor uses a separate path: browser to UNO web API, then the paired UNO proxy to the RetroPie Games service on TCP 55358. Challenge/HMAC exchanges protect registry operations; revision checks prevent stale edits and atomic replacement preserves a previous valid copy. Saving a registry mapping affects the next launch; it does not rewrite the running game's mapping immediately.

Optional native Super Glove Ball path

The supported FCEUmm path consumes the same virtual gamepad as every other game. For native Super Glove Ball input, the authenticated console receiver also publishes a versioned, fixed-size latest-sample record in [/run/virtualglove/native-state](#). The separately built [lr-nestopia-powerglove](#) core maps that file read-only, copies at most one coherent current sample per emulated frame, and adds no queue or smoothing. Invalid, stale, uncalibrated, lost, or wrong-profile samples leave the emulated glove neutral.

Exact-ROM traces now confirm the ten-byte packet boundary, MSB-first reads, native Start, continuous X/Y, signed Z, and open/fist/index packet response. Live full-game play confirms the resulting grab/throw, index-fire, and fist-plus-forward Power Punch actions. A matched same-ROM test confirms that FCEUmm requests only ordinary joypad input while both cores visibly respond to all four directions by frame 3. Stale, uncalibrated, lost, and wrong-profile samples produce a neutral packet.

The shared layer publishes its five-finger closed-hand and index-point decisions explicitly so the core does not reconstruct compound poses from partial finger data. All confirmed Super Glove Ball actions are mapped. The raw roll byte and unobserved button codes remain neutral because the exact ROM has shown no separate action for them; guessing values could create unintended input. Stock Nestopia remains untouched.

RetroPie registers the custom core in its normal secondary-core directory. Recalbox keeps a target-specific core in its persistent share and exposes the separate core plus a current system-list entry through reversible runtime mounts. Its bounded RetroArch process monitor recognizes the actual core command line and authenticates the same native profile identity used by RetroPie.

Batocera keeps the architecture-selected core in persistent [/userdata](#), then uses two reversible overlay mounts to expose only the separately named core and matching info record through Batocera's read-only core paths. The core forces its own Player 1 to the native Power Glove peripheral; other cores and other Nestopia games retain their normal devices. It is enabled only through a Super Glove Ball per-ROM emulator choice after its manifest, corresponding source, ELF identity, and libretro identity are verified and it is load-checked on the console.

The release carries 15 separately targeted Batocera 43.1 cores; it never treats one binary as cross-architecture. Exact registered Super Glove Ball filenames are selected only when no existing per-ROM core choice is present. See [VirtualGlove Input Modes](#).

LaunchBox installs a separately named x86-64 DLL and corresponding source in its per-user VirtualGlove directory. The wrapper verifies the installed DLL before each native launch and publishes the guarded record through a per-user Windows file mapping. A missing or changed DLL selects FCEUmm instead. Ordinary LaunchBox profiles continue through the loopback Network RetroPad; native Super Glove Ball never duplicates those gestures through that route.

The optional project-owned [lr-powerglove-dot](#) core reads the same guarded native-state record but does not emulate a Power Glove packet or load a ROM. A fixed Ports launcher holds a renewable [super_glove_ball/lr-powerglove-dot](#) profile lease while the calibration display is open. The Controller therefore uses the production native X/Y path while the display isolates center, reach, edge clamping, tracking loss, and recovery from game logic.

Interfaces and recovery

Setup's four status markers share the matrix's cached app, console-service, authenticated-response, and independent Networking checks. The read-only `/api/connection-status` endpoint requests bounded background refreshes; no network probe runs on the capture or controller-send path. Unknown or expired results are shown in grey. Reachability and authentication do not establish emulator consumption. See [Setup status](#).

Player operations pass through the bounded same-origin `/api/players` endpoint into the worker. Its tuning lock owns one atomic player/settings/progress file. Generations reject stale writes.

Each player retains a saved calibration. Selection automatically applies the selected player's saved center through the durable restore path, with output paused; players without a saved center require centering.

Version-4 portable backups include the center-box size, personal and effective gesture sensitivity, source software identity, name, and the player's neutral reference. They exclude credentials and Academy progress. Older portable formats are rejected without mutation.

A version-6 player store journals confirmed calibration reuse. The worker writes `calibration.json` before clearing the pending reference and centering gate. Output remains paused until Start controller. The journal resumes after crashes. Older internal stores are reported as unsupported and are not overwritten. Progress writes occur on lesson transitions, not frames.

Hostname resolution for controller sends runs in one background thread with a single cached address. No controller states are retained by that thread. Missing or expired addresses cause the current send to be skipped; later calls use their own newest state.

Host physical Wi-Fi/Ethernet link health is sampled independently by an unprivileged systemd timer, which publishes a small expiring JSON record for the supervisor and fourth Off-mode matrix pixel. Console reachability remains a separate probe.

Build metadata records the source commit and candidate. A generated sketch fingerprint is compiled into firmware and read through Router Bridge in the supervisor, independently of the expected packaged value. Missing readback stays unavailable; this introduces no firmware RPC in the vision worker's frame path.

Interface	Direction	Contract
HTTP 8088	Browser to VirtualGlove Controller	Pages, live status/video, ordinary settings and commands
HTTPS 8443	Browser to VirtualGlove Controller	Secure Setup and pairing workflow
HTTP 8089, loopback	Supervisor/web proxy to worker	Internal status, frame and control requests
UDP 55355	VirtualGlove Controller to console	Signed controller states, session, challenge, and sequence; handshake replies return to the sender socket
UDP 55356	Console to Controller relay to worker	Signed profile requests and acknowledgements
Native-state record	Authenticated console receiver to native cores	<code>/run/virtualglove/native-state</code> on Linux or a per-user mapped file on LaunchBox; guarded latest sample for Super Glove Ball
TCP 55357	Pairing participants	Temporary one-time-code pairing service
TCP 55358	VirtualGlove Controller to console	Paired game-registry and Controller Router services (<code>/registry</code> and <code>/inputs</code>)

Interface	Direction	Contract
Private Unix sockets	App resolver to host Avahi; Recalbox/Batocera receiver to Player 1 merger	Local hostname resolution and bounded local VirtualGlove state delivery
Router Bridge RPC	Linux supervisor to microcontroller	Matrix status/profile/pairing commands

The LAN remains a trust boundary. Controller version 2 uses its own domain-separated HMAC-SHA256 and receiver-issued challenges. It does not encrypt input. Version 1 is neither accepted nor emitted.

Do not describe all links as equivalent secure channels. Pairing and registry exchange have their own protections.

Browser mutations use the existing request-header and Origin checks. See the [Security policy](#) for the full trust model.

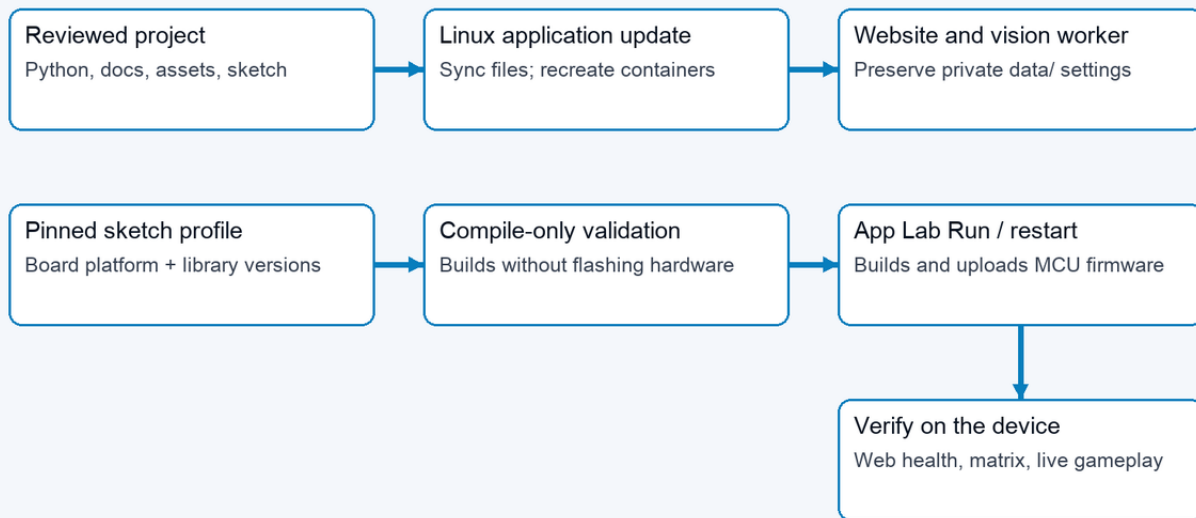
Failure or transition	Implemented response	Interpretation
Hand tracking lost	Engine clears held states after its loss delay	Stops stale recognized actions; camera recovery is separate
Controller packets stop	Receiver releases controls on socket timeout, default 250 ms	A receive timeout, not a measured end-to-end acknowledgement
Hostname or UDP send failure	Sender reports error and throttles retries	Vision and local practice can continue
Camera open/read failure	Worker reports starting/error and retries asynchronously; a sustained failure requests one classified recovery even when USB enumeration remains present. A capability-confirmed hub cycles only the enrolled camera port. Whole-hub fallback is refused when that hub carries networking.	The USB action and camera recovery are separate states. Recovery is confirmed only after the restarted worker receives a frame; lsusb alone does not prove the stream is usable.
Worker exits	Supervisor reports failure and retries	Temporary in-memory Tune state is lost
Tune browser disappears	Six-second lease expires	Preview and recordings discarded; saved pairs retained
Calibration changes	Current Tune recordings/preview invalidated	Record new measurements against the new reference
Shutdown requested	Web action writes fixed request; host systemd helper requests halt	The tested VirtualGlove Controller can restart; not proof that power is safe to remove

A successful UDP send means the local networking call succeeded. It does not prove the receiver applied a state or the game accepted it. Diagnose in stages: hand detected, measured values, recognized/held action, delivery gate, sender error, receiver/gamepad state, then emulator/game mapping.

Setup reports the configured console name separately from the active authenticated input address. Its downloadable version-1 [virtualglove-system-report](#) is deliberately allowlisted rather than a dump of internal state. It includes software/firmware identity, camera reader and rate, controller/profile state, and connection-check results. It excludes frames, pairing keys, player/calibration data, ROM names, and network addresses.

Build and deployment

07 / Linux application and matrix firmware update separately



Installing a platform alone changes neither application behaviour nor the running matrix firmware.

Separate Linux application and microcontroller firmware deployment paths

The versioned [install-uno-q.sh](#), [install-retropie.sh](#), [install-recalbox.sh](#), and [install-batocera.sh](#) entry points download matching packages and call the shared host installer. LaunchBox uses its extracted per-user PowerShell installer.

The UNO route uses App Lab CLI to build/upload the sketch and start the app; it installs both startup and fixed-purpose shutdown/camera-recovery helpers. Each console route installs its receiver and platform-specific game-session integration in the platform's supported persistent location, then checks emulator and registered-game configuration.

There are two deployable parts. Python, website, documentation, assets, and service support run on Linux. The **Arduino sketch** is the microcontroller source code; its compiled and installed version is the **matrix firmware**. That firmware drives the LED matrix and handles Router Bridge commands.

The Wi-Fi deployment script synchronizes Linux application files and recreates containers; it does not upload matrix firmware. A documentation-only sync can serve new Markdown and PDFs without restarting the application, provided Python route registration has not changed.

The Arduino sketch currently depends on the Arduino Zephyr platform **1.0.0** for [arduino:zephyr:unoq](#). Zephyr is the current platform dependency, rather than the name of the VirtualGlove component. The build configuration also pins:

- [Arduino_RouterBridge 0.4.3](#)
- [Arduino_RPClite 0.3.0](#)
- [ArxContainer 0.7.0](#)
- [ArxTypeTraits 0.3.2](#)
- [DebugLog 0.8.4](#)

- MsgPack **0.4.2**

The verified platform supplies Arduino_LED_Matrix **0.1.3**. Retain the complete project [sketch/sketch.yaml](#) when synchronizing with App Lab.

Installing that platform makes build tools available. Compile-only validation builds against it but does not flash hardware. App Lab **Run**, or its supported app-restart command, compiles the Arduino sketch and uploads the matrix firmware.

Back up the installed source and firmware cache, verify compilation, upload, then check application health, bridge response, physical matrix appearance, and actual controls. Keep private settings intact. Detailed commands are in the [Installation Guide](#).

Documentation has an editable Markdown source, generated diagrams, built-in Help rendering, and a PDF edition. [scripts/build-architecture-diagrams.py](#) regenerates these nine figures. [scripts/build-docs-pdf.py](#) generates the PDF set. The Help and package allowlists explicitly include this architecture guide. The local quick reference remains excluded from public deployment.

Implementation map

Paths below are relative to the project root. This map identifies responsibility; it does not claim every path has been independently security-audited.

Responsibility	Start reading here
Supervisor, worker launch, matrix ownership	python/main.py
Camera lifecycle and frame-to-send loop	src/virtualglove/vision_app.py , realtime.py
Capture selection, Kiyo controls, and landmark measurements	src/virtualglove/camera.py , kiyo_camera.py , tracker.py
Current native X/Y implementation and diagnostics	gesture.py , realtime.py
Observation/state data objects	src/virtualglove/model.py
Calibration, thresholds, held gestures, mappings	src/virtualglove/gesture.py
Recording, suggestions, previews, persistence	src/virtualglove/tuning.py
Public HTTP routing and worker proxy	src/virtualglove/control_server.py
Shared page shell and maintained browser modules	web_common.py , dashboard_web.py , academy_web.py , games_web.py , tuning_web.py , setup_web.py , player_web.py
Worker requests, status, practice leases	src/virtualglove/debug_server.py
Controller packets, console output, and Players 1-4 routing	src/virtualglove/transport.py , controller_protocol.py , receiver.py , controller_router.py ; merged_gamepad.py retains migration primitives
Optional RetroPie I-PAC/8BitDo/VirtualGlove cabinet merger	retropie/arcade-cabinet-merger/

Responsibility	Start reading here
Profile requests, launch hooks, UDP relay	src/virtualglove/profile_control.py , retropie_hook.py , scripts/profile-relay.py
Paired Games editing	src/virtualglove/game_registry.py
Pairing and hostname resolution	src/virtualglove/pairing.py , python/ssh_pair.py , src/virtualglove/resolver.py
Matrix translation and firmware	src/virtualglove/matrix.py , sketch/sketch.ino
App services and installation	app.yaml , bricks/local/ , scripts/setup-machine.py
Help and printable guides	src/virtualglove/help_content.py , scripts/build-docs-pdf.py

Validation boundaries

Automated tests establish data contracts, safety behavior, configuration persistence, and deterministic input handling. Hardware checks establish camera, matrix, bridge, receiver, and emulator integration. Live gameplay establishes that recognized movement and gestures remain usable as one end-to-end system. No one category substitutes for the others.

Before releasing recognition changes, exercise optional hand setup and individual tuning without setup; V-sign and thumbs-up with different curl ranges; incorrect extended fingers; incomplete releases; tracking loss; insufficient/overlapping samples; calibration changes; preview expiry; persistence; and reset. Confirm that feedback agrees with recognition and output remains paused throughout Glove Academy/Tune. Complete live camera and gameplay tests before describing a reduced recording count as validated for other users.

Matrix during startup

The Arduino sketch shows an hourglass before its blocking Router Bridge setup. A dedicated display task owns subsequent framebuffer writes and keeps startup feedback moving independently of Linux and Python initialization.

The main sketch task registers the bridge endpoints; those endpoints update requested status/profile values, and the display task renders them. If the display-task stack allocation fails, the first hourglass stays visible during setup and the normal sketch loop takes over rendering afterward. This task currently uses the Zephyr API supplied by the Arduino sketch platform.

Python requests loading before importing the web controls, then forwards normal worker status. The hourglass indicates activity, not measured completion. It does not replace the protected system boot display.

The optional host user service [virtualglove-early-start.service](#) releases the installed sketch earlier using the loader release flag, after checking the selected app and sketch samples. It never resets, halts, or flashes the sketch. This brings the existing hourglass forward while App Lab continues starting. Failure falls back to normal App Lab startup; the cold-boot trial was confirmed on the physical board.

Idle display preferences

The supervisor passes the persisted [matrix_attract](#) setting to the sketch through [set_virtualglove_attract\(mode, connections\)](#). Only [PG_GESTURES_IDLE](#) consumes it; there is no global brightness change. In Off mode a bounded background probe checks TCP reachability and authenticates the existing RetroPie Games service. The supervisor publishes cached indicator bits; capture, recognition, transport, T/L displays, and game-state paths are unchanged.

Signed controller session lifecycle

After either supported pairing method installs the shared token, the Controller sends a new signed hello to UDP 55355 and requires a matching receiver challenge before reporting success. This verifies that RetroPie is running the receiver and accepts the token just written. It does not arm controller delivery or prove that RetroArch, an emulator, or a game consumed input.

The nonblocking sender emits a signed hello with random session and request identifiers. RetroPie replies with a fresh random 128-bit challenge; only a signed reply matching the sender's current request, session, and configured receiver port is accepted.

On Linux, receiver replies preserve the destination address and receiving interface using `IP_PKTINFO`, so Ethernet/Wi-Fi multihoming works through container NAT. The sender also permits a different source address when HMAC, request, session, and port match.

A valid state activates that challenge. Activation invalidates every older active and pending challenge; subsequent states require increasing sequence numbers. A replayed hello can obtain a new challenge but cannot supply an authenticated state for it. Receiver restarts discard all challenges, so recorded traffic from a previous process cannot activate input.

At most eight pending handshakes are retained, for three seconds each. No input state is retained while negotiating. Hellos repeat every 250 milliseconds before the first challenge, then once per second to recover a receiver restart.

The sender reads at most eight replies per update without blocking and sends only that update's state. Periodic handshake traffic does not reset the receiver's input-release deadline. Both native-state publication and `uinput` remain behind the same accepted-state check; the core and recognition paths are unchanged.

Dashboard, Academy, Games, personalization, Play, and Setup each import their maintained page from the owning module. The unused compatibility re-export and duplicate worker homepage have been removed.

For a guided symptom check, see [Troubleshooting by symptom](#).